Ericsson PE Node WFD-PE-02 Standard Operating Procedure for Minor Severity Routing Protocol Flap ALARM_9577

Node: WFD-PE-02 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_9577

Description

The Ericsson PE node WFD-PE-02 at the Whitefield POP has encountered a minor severity Routing Protocol Flap ALARM_9577, primarily due to BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This failure domain occurs due to the inherent complexities of the routing protocol interactions between the PE node, the Route Reflector (RR) cluster, and the autonomous system border routers (ASBRs). The PE node's primary role is to aggregate and redistribute routing information between the customer-facing network and the provider core network. However, due to the dynamic nature of BGP, OSPF, and ISIS, the PE node may experience routing protocol flaps, leading to temporary loss of adjacency and flooding of LSPs. This can be exacerbated by factors such as configuration inconsistencies, hardware or software issues, or network instability. To mitigate this issue, it is essential to perform a thorough diagnostic and troubleshooting process to identify the root cause and implement the necessary corrective actions.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Whitefield POP to facilitate access and troubleshooting.

- Verify the status of redundant links and ensure they are operational.

- Review the PE node's configuration and ensure that it is consistent with the current network topology and routing policies.

Potential Root Causes

Cause	Probability
BGP route churn due to configuration inconsistencies or network instability	High
OSPF adjacency loss due to hardware or software issues	Medium
ISIS LSP flooding due to network instability or configuration errors	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version 1 BGP scanner interval is 60 sec Status codes: s-suppressed, d-damped, h-hold, *-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i - IGP, e - EGP, ? - incomplete, Description: This command displays a summary of the BGP routing table, including the BGP table version, main routing table version, and BGP scanner interval. It also provides status codes and origin codes for each entry in the table.
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```
Command: show ip ospf neighbor, Expected Output: OSPF Process ID: 1 Router ID: 10.10.10.1 Neighbor Count: 1 Neighbors: Neighbor ID: 10.10.10.2 State: Full Address: 10.10.10.2 Interface: Ethernet0/0, Description: This command displays information about OSPF neighbors, including the neighbor's router ID, state, address, and interface.

Command: show isis database, Expected Output: ISIS Level 1 Database: L1 LSP ID: 10.10.10.1 Sequence Number: 0x00000001 Checksum: 0x12345678 Length: 20 ..., Description: This command displays the ISIS database, including L1 LSPs, sequence numbers, checksums, and lengths.
```

Resolution Steps

Step 1: Step: Verify BGP routing table consistency, Command: show ip bgp summary, Description: Verify that the BGP routing table is consistent and free of errors.

Step 2: Step: Check OSPF adjacency, Command: show ip ospf neighbor, Description: Verify that the OSPF adjacency is stable and that all neighbors are reachable.

Step 3: Step: Flush ISIS LSPs, Command: clear isis database, Description: Flush the ISIS LSPs to reset the ISIS database and re-converge the network.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version 1 BGP scanner interval is 60 sec Status codes: s-suppressed, d-damped, h-hold, *-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i - IGP, e - EGP, ? - incomplete, Description: Verify that the BGP routing table is consistent and free of errors.

Command: show ip ospf neighbor, Expected Output: OSPF Process ID: 1 Router ID: 10.10.10.1 Neighbor Count: 1 Neighbors: Neighbor ID: 10.10.10.2 State: Full Address: 10.10.10.2 Interface: Ethernet0/0, Description: Verify that the OSPF adjacency is stable and that all neighbors are reachable.

Command: show isis database, Expected Output: ISIS Level 1 Database: L1 LSP ID: 10.10.10.1 Sequence Number: 0x00000001 Checksum: 0x12345678 Length: 20 ..., Description: Verify that the ISIS LSPs are stable and free of errors.
```

Roll-back Procedures

- Step: Restore previous configuration, Command: load merge /config/previous_config, Description: Restore the previous configuration to revert any changes made during the troubleshooting and resolution process.

- Step: Revert ISIS database, Command: clear isis database revert, Description: Revert the ISIS database to its previous state to undo any changes made during the troubleshooting and resolution process.